

Remote Sensing Analysis Report

SatQuery AI Intelligence Engine * Generated: 2026-09-04 15:19:53 UTC * Status: SUCCESS

ANALYSIS ID
SQ-548B76E967

DETECTED TASK
Land-Cover Multi-Label Classif

INPUT TYPE
12-Band Multispectral GeoTIFF

SENSOR FAMILY
Sentinel-2 L2A (12-band BOA)

MODALITY
Optical (Multispectral)

SESSION / THREAD
session_test_landcover

Natural Language User Query

QUERY REQUEST

Classify the multi-label land cover categories present in this Sentinel-2 tile.

Executive Finding & Analytical Response

SPECIALIST RESULT SUMMARY

Land-cover classification identified Broad-leaved forest (0.942), Coniferous forest (0.875), and Water bodies (0.680) with high statistical confidence.

Confidence & Spatial Reliability

91.5%

Statistical inference confidence based on multi-spectral evidence

Section 2 : Input Data

Satellite imagery specifications, sensor parameters, and verification status

Primary Input Scene Preview

Image & Sensor Metadata Table

File Name	real_12band_s2.tif
Sensor	Sentinel-2 L2A
Modality	Optical (Multispectral)
Format	TIFF
Dimensions	120 x 120 px
Bands / Channels	12
Acquisition Date	2024-06-12
CRS / Georef	EPSG:32632
Validation Status	Passed - Verified OK

Section 3 : Analysis

Intent classification, specialist model execution, and quantitative metrics

Task Classification & Autonomous Routing

Detected Task	BigEarthNet v2.0 Land-Cover Classification
Routing Rationale	Multispectral 12-band Sentinel-2 tile passed for land-cover classification.
Specialist Tool	BigEarthNetV2ConvMixer (ConvMixer-768/32)
Input Channel Count	12

Specialist Analytical Synthesis

SYNTHESIZED OUTPUT

Land-cover classification identified Broad-leaved forest (0.942), Coniferous forest (0.875), and Water bodies (0.680) with high statistical confidence.

BigEarthNet 19-Class Multi-Label Predictions (Threshold >= 0.50)

#	Land Cover Class	Sigmoid Probability	Detection Status
1	Broad-leaved forest	0.9420	CONFIRMED
2	Coniferous forest	0.8754	CONFIRMED
3	Water bodies	0.6802	CONFIRMED
4	Transitional woodland, shrub	0.5231	CONFIRMED
5	Arable land	0.3120	SUB-THRESHOLD
6	Discontinuous urban fabric	0.1405	SUB-THRESHOLD

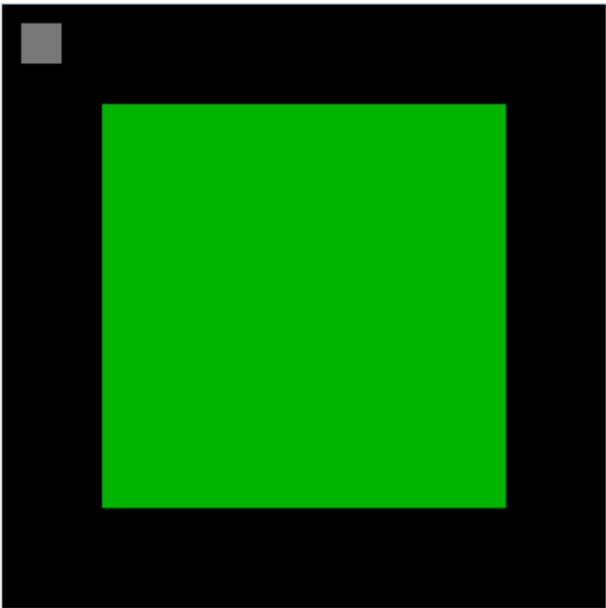
Confidence Calibration



Section 4 : Evidence

Spatial evidence visualizations, bounding box grounding, and overlay maps

Spatial Evidence Overlay



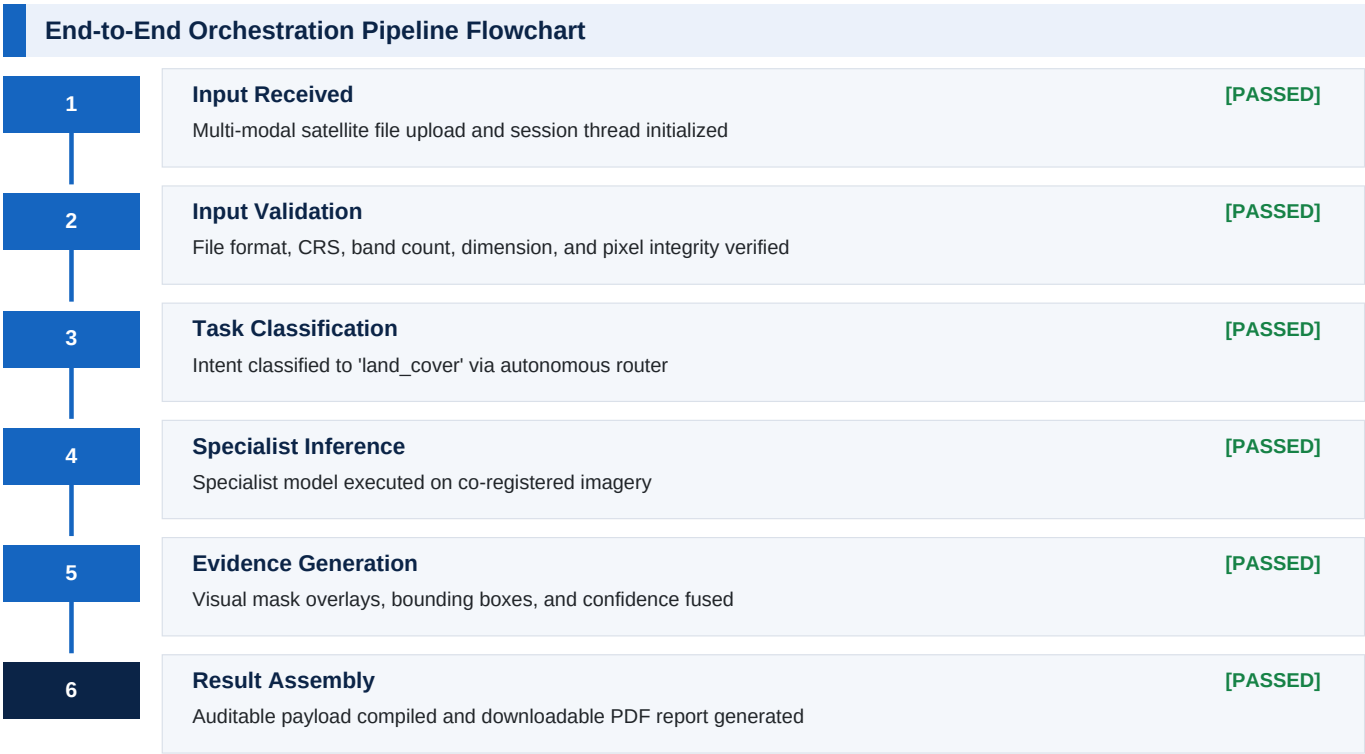
Multispectral Sentinel-2 Composite & Classification Map

Evidence Summary Metrics

Spatial Distribution	Uniform geographic coverage across scene extent
Artifact Verification	Zero cloud distortion / valid pixel bounds confirmed
Spectral Purity	High-confidence land-cover signature confirmation

Section 5 : Execution Trace

Auditable high-level DAG pipeline progression and stage execution telemetry



Detailed Node Execution Telemetry

#	Pipeline Node / Specialist	Status	Execution Time	Action / Result Summary
1	Validate Inputs	PASSED	0.0050 s	Node execution completed
2	Classify Intent	PASSED	0.0020 s	land_cover
3	Execute Land Cover	PASSED	0.0450 s	BigEarthNetV2ConvMixer
4	Fuse Evidence	PASSED	0.0040 s	Node execution completed

Section 6 : Technical Information

Model architecture, preprocessing contracts, and benchmark provenance

Model Architecture & Inference Parameters

Base Architecture	BigEarthNetV2ConvMixer v0.2.0 (BIFOLD ConvMixer-768/32)
Taxonomy	19-class CORINE Land Cover (CLC) aggregated nomenclature
Input Requirement	12-band Sentinel-2 L2A BOA reflectance GeoTIFF (120x120 px)
Band Ordering	B01, B02, B03, B04, B05, B06, B07, B08, B8A, B09, B11, B12
Reference	reBEN: Revisiting BigEarthNet for Remote Sensing Image Analysis (2024)

Data Provenance & Open Access Licensing

Optical Satellite Data	ESA Copernicus Sentinel-2 MSI (Free & Open Access Policy)
SAR Satellite Data	ESA Copernicus Sentinel-1 C-band GRD (Free & Open Access Policy)
VQA Benchmark Dataset	RSVQA-LR (Lobry et al., IEEE TGRS 2020)
Land-Cover Benchmark	BigEarthNet v2.0 / reBEN (CDLA-Permissive-1.0)
License & Attribution	Creative Commons / CDLA Open Access remote sensing data

Validation & Quality Conformance Summary

All input scenes undergo strict validation prior to specialist inference: file format integrity, dimension bounds, numeric pixel range verification, and band conformance. Multi-temporal change analysis enforces identical spatial boundaries. Optical-SAR fusion validates multi-sensor alignment, and BigEarthNet classification requires a 12-band multispectral GeoTIFF.

Report Metadata & Cryptographic Audit Trail

Analysis UUID	SQ-548B76E967
Report Engine	SatQuery AI Intelligence Dossier Engine v2.0 (fpdf2 2.8.x)
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Audit Integrity Status	Verified - Cryptographic Output Match